1D Flood Model of a Small Creek (HEC-RAS)

A short demonstration of the HEC-RAS flood-mapping workflow. Indicative results only.

1. What we did

We built a 1D steady-flow HEC-RAS model of a small mountain creek (Piru Creek, Ventura County, USA), over a reach about 1.655 km long. We ran three scenario flows and mapped the flooded extents. The aim was to show the workflow (terrain, geometry, flow, run, map), not to produce a design flood map.

2. Data used

Terrain: USGS 3DEP 1 m DEM, clipped to the reach plus a 300 m buffer

Imagery and basemap: Esri World Imagery; OpenTopoMap (© OpenStreetMap contributors)

Software: HEC-RAS 7.0 (RAS Mapper), QGIS

3. Method

1. **Slope.** Elevations were sampled from the DEM every 20 m along a 1,655 m reach line, and a line was fitted to the profile. This gave S = 0.0135 m/m, used for the Normal Depth boundary.
2. **Terrain.** The clipped DEM was added to RAS Mapper with the project's coordinate system.
3. **Geometry.** The river centreline and cross sections were drawn on the terrain, and elevations were cut from the DEM. Manning's n was set to 0.050 (left bank), 0.035 (channel) and 0.050 (right bank).
4. **Flows.** Three scenario flows were entered at the upstream end: A = 5, B = 20 and C = 60 m^3/s. Both boundaries use Normal Depth, slope 0.0135.
5. **Run and maps.** The steady-flow plan ran with no warnings or errors. Depth and inundation-boundary layers were created for each scenario, and the boundaries were mapped over the basemap in QGIS.

4. What we saw

Results at the downstream section (River / main / RS -1):

	A	B	C
Flow (m^3/s)	5	20	60
Water surface (m)	404.67	405.37	406.29
Max channel depth (m)	0.56	1.26	2.18
Top width (m)	5.4	7.2	13.5
Mean velocity (m/s)	1.90	2.87	3.48

- Depth, width and velocity all rise with flow, as expected.
- Flow is close to critical at every scenario. The water surface sits only 0.03 to 0.06 m above critical depth, and the Froude number (calculated by hand) rises from about 0.87 to 0.98. This is normal for a steep creek ($S \approx 0.0135$).
- The energy slope matches the boundary slope, so the Normal Depth boundary behaves as intended.

5. Limitations

- The flows are **assumed scenario values**, not a flood-frequency analysis, so they are not return periods.
- Manning's n values are assumed.
- The cross sections are narrow (about 18 m in the one we inspected), and water reached the end of a section in an earlier plot. The mapped flooded widths are therefore limited by the section lengths and are lower bounds.
- No bank lines were drawn, so the whole section is treated as channel.
- The reach is a steep, narrow canyon, so the three extents may look similar.
- Only the downstream section's values are reported here. It is also the section most affected by the boundary condition.
- No sensitivity analysis was done. Bridges, culverts and structures are not modelled.

Conclusion: the model runs cleanly and demonstrates the HEC-RAS workflow. The results are indicative and are not suitable for flood hazard assessment or design.